A polynomial with a root in $(0, 1)$

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Theorem 1. *If*

$$C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0,$$

where $C_0, \ldots, C_n$ *are real constants, then the equation*

$$C_0 + C_1 x + \cdots + C_{n-1} x^{n-1} + C_n x^n = 0$$

has at least one real root between 0 *and* 1.

Proof. Write the hypothesis as

$$\sum_{k=0}^{n} \frac{C_k}{k+1} = 0,$$

and define

$$F(y) = \sum_{k=0}^{n} \frac{C_k y^{k+1}}{k+1} = C_0 y + \frac{C_1}{2} y^2 + \cdots + \frac{C_n}{n+1} y^{n+1}.$$

We show that the derivative of F is the given polynomial.

Fix k and x. As a real number, $k + 1 \neq 0$, indeed $k + 1 > 0$ since $k \geq 0$. The function $y \mapsto y^{k+1}$ has derivative $(k+1)x^k$ at x. Hence, multiplying by the constant C_k and dividing by the constant $k + 1$, the function

$$y \mapsto \frac{C_k y^{k+1}}{k+1}$$

has derivative

$$\frac{C_k (k+1) x^k}{k+1} = C_k x^k$$

at x. Differentiating a finite sum term by term, which follows by repeated application of the sum rule, we obtain that F is differentiable everywhere with derivative

$$F'(x) = \sum_{k=0}^{n} C_k x^k = C_0 + C_1 x + \cdots + C_{n-1} x^{n-1} + C_n x^n.$$

In particular, F is continuous on $[0, 1]$.

At 0, each term of F vanishes because $0^{k+1} = 0$, so

$$F(0) = \sum_{k=0}^{n} \frac{C_k \cdot 0^{k+1}}{k+1} = 0.$$

At 1, we have $1^{k+1} = 1$, so

$$F(1) = \sum_{k=0}^{n} \frac{C_k \cdot 1^{k+1}}{k+1} = \sum_{k=0}^{n} \frac{C_k}{k+1} = C_0 + \frac{C_1}{2} + \cdots + \frac{C_n}{n+1} = 0$$

by hypothesis. Thus $F(0) = F(1)$.

By Rolle's theorem, there exists $c \in (0, 1)$ such that $F'(c) = 0$. But $F'(c)$ is the value of the given polynomial at c, so

$$C_0 + C_1 c + \cdots + C_{n-1} c^{n-1} + C_n c^n = 0.$$

Thus c is a real root of the equation between 0 and 1. □